PRACTICE SHEET

1. What is the value of x that satisfies the equation $\cos^{-1} x = 2 \sin^{-1} x$?

(a) 1/2 (b) –1
(c) 1 (d) –1/2

2. What is the value of $\sin^{-1}\left(\sin\frac{2\pi}{3}\right)$?

(a) $-\pi/3$ (b) $2\pi/3$
(c) $-2\pi/3$ (d) $\pi/3$

3. What are the values of (x, y) satisfying the simultaneous equations $\sin^{-1} x + \sin^{-1} y = \frac{2\pi}{3}$ and $\cos^{-1} x - \cos^{-1} y = \frac{\pi}{3}$ =?

(a) (0, 1) (b) $(\frac{1}{2},1)$
(c) $(1,\frac{1}{2})$ (d) $(\frac{\sqrt{3}}{2},1)$

4. What is $\sin[\cot^{-1}\{\cos(\tan^{-1}x)\}]$ where x>0, equal to?

(a) $\sqrt{\frac{(x^2+1)}{(x^2+2)}}$ (b) $\sqrt{\frac{(x^2+2)}{(x^2+1)}}$
(c) $\frac{(x^2+1)}{(x^2+2)}$ (d) $\frac{(x^2+2)}{(x^2+1)}$

(a) $x = -\frac{1}{2}$ (b) x =1
(c) $x = \frac{1}{2}$ (d) $x = \frac{\sqrt{3}}{2}$

10. If $\sin^{-1} x + \sin^{-1} y = \pi/2$ and $\cos^{-1} x - \cos^{-1} y = 0$, then values x and y are respectively

(a) $\frac{1}{\sqrt{2}},-\frac{1}{\sqrt{2}}$ (b) $\frac{1}{2},\frac{1}{2}$
(c) $\frac{1}{2},-\frac{1}{2}$ (d) $\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}$

11. The formula $\sin^{-1}\left\{2x\left(1-x^2\right)\right\} = 2\sin^{-1} x$ is true for all values of x lying in the interval

(a) [–1, 1] (b) [0, 1]
(c) [–1, 0] (d) $[-1/\sqrt{2},1/\sqrt{2}]$

12. If $\sin^{-1}\left(\frac{2a}{1+a^2}\right)-\cos^{-1}\left(\frac{1-b^2}{1+b^2}\right)=\tan^{-1}\left(\frac{2x}{1-x^2}\right)$, then what is the value of x?

(a) a / b (b) ab
(c) b / a (d) $\frac{a-b}{1+ab}$

13. If $\sin^{-1} x + \cot^{-1}(1/2) = \pi/2$, then what is the value for x?

(a) 0 (b) $1/\sqrt{5}$